*What will be the output of the following programs*

*1.*

```
// This program uses variables and literals.
#include <iostream>
using namespace std;
int main()
{
int little;
int big;
little = 2;
big = 2000;
cout << "The little number is " << little << endl;
cout << "The big number is " << big << endl;
return 0;
}
```

Output:

The little number is 2
The big number is 2000

2.

```
// This program calculates the
// circumference of a circle.
#include <iostream>
using namespace std;
int main()
{
   const double PI = 3.14159;
   const double DIAMETER = 10.0;
   double circumference;
   circumference = PI * DIAMETER;
   cout << "The circumference is: "
        << circumference << endl;
   return 0;
}
```

Output:

The circumference is: 31.4159

Short Answer

**1.** How many operands does each of the following types of operators require?

_____1_____ Unary
_____2_____ Binary
_____3_____ Ternary

**2.** How may the double variables temp, weight, and age be defined in one statement?

Answer: double temp, weight, age;

**3.** How may the int variables months, days, and years be defined in one statement, with months initialized to 2 and years initialized to 3?

Answer: int months = 2, days, years = 3;

**4.** Write assignment statements that perform the following operations with the variables a, b, and c:

A. Adds 2 to a and stores the result in b. b = a + 2;

B. Multiplies b by 4 and stores the result in a. a = b * 4;
C. Divides a by 3.14 and stores the result in b. b = a / 3.14;
D. Subtracts 8 from b and stores the result in a. a = b - 8;
E. Stores the value 27 in a. a = 27;
F. Stores the character 'K' in c. c = 'K';
G. Stores the ASCII code for 'B' in c. c = (int)'B';

**5.** Is the following comment written using single-line or multi-line comment symbols?
*/* This program was written by M. A. Codewriter*/*
Answer: Multi-line comment symbols.

**6.** Is the following comment written using single-line or multi-line comment symbols?
*// This program was written by M. A. Codewriter*
Answer: single-line comment symbols.

**7.** Modify the following program so it prints two blank lines between each line of text.

```
#include <iostream>
using namespace std;
int main()
{
cout << "Two mandolins like creatures in the";
cout << "dark";
cout << "Creating the agony of ecstasy.";
cout << " - George Barker";
return 0;
}
```

Answer:

```
#include <iostream>
using namespace std;

int main()
{
   cout << "Two mandolins like creatures in the\n\n\n";
   cout << "dark\n\n\n";
   cout << "Creating the agony of ecstasy.\n\n\n";
   cout << " - George Barker";
   return 0;
}
```

**8.** What will the following programs print on the screen?

A.
```
#include <iostream>
using namespace std;
int main()
{
int freeze = 32, boil = 212;
freeze = 0;
boil = 100;
cout << freeze << endl << boil << endl;
return 0;
}
```

Answer:
0
100

B. **#include <iostream>**
**using namespace** std;
**int main**()
{
**int** x = 0, y = 2;
x = y * 4;
cout << x << endl << y << endl;
**return** 0;
}
Answer:
8
2

C. **#include <iostream>**
**using namespace** std;
**int main**()
{
cout << "This app tracks your";
cout << "daily\nhabits";
cout << "\nand helps you\nstay\n";
cout << "motivated.";
**return** 0;
}
Answer:
This app tracks yourdaily
habits
and helps you
stay
motivated.
D. **#include <iostream>**
**using namespace** std;
**int main**()
{
cout << "Be careful\n";
cout << "This might/n be a trick ";
cout << "question\n";
**return** 0;
}
Answer:
Be careful
This might/n be a trick question

E. **#include <iostream>**
**using namespace** std;
**int main**()
{
**int** a, x = 23;
a = x % 2;
cout << x << endl << a << endl;

```
return 0;
}
```

Answer:
23
1

Multiple Choice

**9.** Every complete statement ends with a(n) _____.
- **a.** period
- **b.** # symbol
- **c.** semicolon
- **d.** ending brace

**10.** Which of the following statements is correct?
- **a.** #include (iostream)
- **b.** #include {iostream}
- **c.** #include <iostream>
- **d.** #include [iostream]
- **e.** All of the above

**11.** Every C++ program must have a _____.
- **a.** cout statement
- **b.** function main
- **c.** #include statement
- **d.** All of the above

**12.** Preprocessor directives begin with _____.
- **a.** #
- **b.** !
- **c.** <
- **d.** *
- **e.** None of the above

**13.** The following data

72
'A'
"Hello World"
2.8712
are all examples of _____.

- **a.** variables
- **b.** literals or constants
- **c.** strings
- **d.** none of the above

**14.** A group of statements, such as the contents of a function, is enclosed in _____.
- **a.** braces {}
- **b.** parentheses ()
- **c.** brackets <>
- **d.** all of the above will do

**15.** Which of the following are *not* valid assignment statements? (Select all that apply.)
- **a.** total = 9;
- **b.** 72 = amount;
- **c.** profit = 129
- **d.** letter = 'W';

**16.** Which of the following are not valid cout statements? (Select all that apply.)

**a.** cout << "Hello World";
**b.** cout << "Have a nice day"\n;
**c.** cout < value;
**d.** cout << Programming is great fun;

**17.** Assume w = 5, x = 4, y = 8, and z = 2. What value will be stored in result in each of
the following statements?
**a.** result = x + y; 12
**b.** result = z * 2; 4
**c.** result = y / x; 2
**d.** result = y − z; 6
**e.** result = w % 2; 1

**18.** How would each of the following numbers be represented in E notation?
**a.** 3.287 × 106: 3287e3
**b.** −978.65 × 1012: -97865e10
**c.** 7.65491 × 10−3: 765491e-8
**d.** −58710.23 × 10−4: −5871023e-6

**19.** The negation operator is _____.
**a.** unary
**b.** binary
**c.** ternary
**d.** none of the above

**20.** A(n) _____ is like a variable, but its value is read-only and cannot be changed during the program's execution.
**a.** secure variable
**b.** uninitialized variable
**c.** named constant
**d.** locked variable

**21.** When do preprocessor directives execute?
**a.** Before the compiler compiles your program
**b.** After the compiler compiles your program
**c.** At the same time as the compiler compiles your program
**d.** None of the above

## True or False

**22.** T F A variable must be defined before it can be used.
**23.** T F Variable names may begin with a number.
**24.** T F Variable names may be up to 31 characters long.
**25.** T F A left brace in a C++ program should always be followed by a right brace later in the program.
**26.** T F You cannot initialize a named constant that is declared with the const modifier.

## Algorithm Workbench

**27.** Convert the following *pseudocode* to C++ code. Be sure to define the appropriate variables.

Store 20 in the *speed* variable.
Store 10 in the *time* variable.
Multiply *speed* by *time* and store the result in the *distance* variable.
Display the contents of the *distance* variable.

Answer:

```
#include <iostream>
using namespace std;

int main()
{
   int speed = 20, time = 10;
   int distance = speed * time;
   cout << distance;

   return 0;
}
```

**28.** Convert the following *pseudocode* to C++ code. Be sure to define the appropriate variables.

Store 172.5 in the *force* variable.
Store 27.5 in the *area* variable.
Divide *area* by *force* and store the result in the *pressure* variable.
Display the contents of the *pressure* variable.

Answer:

```
#include <iostream>
using namespace std;

int main()
{
   float force = 172.5, area = 27.5;
   float pressure = force / area;
   cout << pressure;

   return 0;
}
```

## Find the Errors

**29.** There are a number of syntax errors in the following program. Locate as many as you can.

```
*/ What's wrong with this program? /*          (correct: /* */)
#include iostream                              (correct: <iostream>)
using namespace std;
int main();              (adding ; will close the main function. Should be without ;)
}                        (correct: {, as the written is for closing)
int a, b, c \\ Three integers          (no semicolon)
a = 3                                  ( ~ ~ ~)
```

```
b = 4                                   ( ~ ~ ~)
c = a + b                               ( ~ ~ ~)
Cout < "The value of c is %d" < C; (correct: cout << … << c; %d doesn't work but with printf())
return 0;
{                                       (correct: }, the written is for opening)
```